

Online Appendix to

“Constructed Exposure Measures and Statement-Specific Robustness: Evidence
from Early-Career Employment”

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A Exposure architectures and measurement lineage

The six architectures differ before any labor outcome enters. Three versions of the ability-based index alter the weighting used to aggregate occupation–ability links. Three task-based measures use the direct-exposure share (alpha), the direct plus complement-software share (beta), and the broad direct-or-complement share. The source files, transformations, and occupation-level checks are recorded in the replication ledger. Figure 1 summarizes the construction paths.

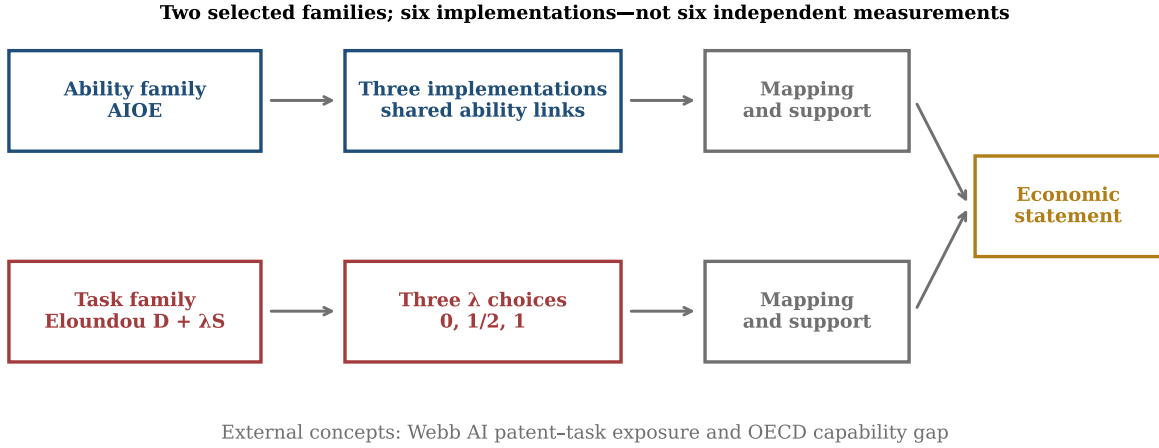


FIGURE 1. Measurement lineage across selected and external AI-exposure architectures. Each path ends in an occupation-level score before merger to the employment data. Ability-based, task-based, patent-overlap, and capability-gap inputs represent distinct treatment constructions, not alternative labels for observed adoption.

TABLE 1. Employment-weighted correlations among exposure architectures

	AIOE-A	AIOE-D	AIOE-S	Alpha	Beta	Broad
<i>Panel A. Pearson correlations</i>						
AIOE administrative	1.000	0.984	0.996	0.295	0.833	0.890
AIOE ability-direct	0.984	1.000	0.981	0.276	0.821	0.883
AIOE source-weighted	0.996	0.981	1.000	0.293	0.832	0.890
Alpha	0.295	0.276	0.293	1.000	0.643	0.346
Beta	0.833	0.821	0.832	0.643	1.000	0.941
Broad	0.890	0.883	0.890	0.346	0.941	1.000
<i>Panel B. Average-rank correlations</i>						
AIOE administrative	1.000	0.987	0.993	0.274	0.844	0.874
AIOE ability-direct	0.987	1.000	0.982	0.258	0.845	0.880
AIOE source-weighted	0.993	0.982	1.000	0.269	0.843	0.874
Alpha	0.274	0.258	0.269	1.000	0.567	0.369
Beta	0.844	0.845	0.843	0.567	1.000	0.961
Broad	0.874	0.880	0.874	0.369	0.961	1.000

Notes: Frozen treatment-side reference sample of 463 occupations, weighted by pre-period employment. Panel B is the weighted Pearson correlation of deterministic average ranks. These are descriptive treatment-side comparisons.

TABLE 2. Employment-weighted correlations with occupational characteristics

Architecture	Cognitive	Manual	RTI	Educ.	Wage	Remote	STEM	Computer
AIOE administrative	.653	-.936	.161	.688	.611	.741	.249	.849
AIOE ability-direct	.689	-.913	.152	.708	.640	.746	.242	.847
AIOE source-weighted	.660	-.939	.148	.700	.617	.754	.249	.854
Alpha	-.032	-.266	.217	-.034	.011	.200	.436	.304
Beta	.478	-.758	.169	.425	.478	.589	.371	.797
Broad	.598	-.810	.108	.530	.600	.618	.253	.833

Notes: Descriptive Pearson correlations using frozen pre-period employment weights. RTI is routine-task intensity. These characteristics are not validation labels for true AI exposure.

For the task family, the raw primitives satisfy $X_\alpha = D$, $X_\beta = D + S/2$, and $X_\gamma = D + S$. On strict common support, the maximum absolute gap between raw beta and $(\alpha + \gamma)/2$ is 7.0×10^{-13} . Mapping and aggregation preserve the identity. Separate employment-weighted standardization changes the relative location and scale and creates a maximum numerical midpoint gap of 0.5112; it does not create a third primitive.

The external-architecture admission rule was written before fitting external outcomes. Webb AI and the reversed OECD capability gap meet the declared construct, public-source, non-title mapping, full-component, distinct-cut, and 80-percent coverage criteria. Webb software remains a computerization control; its source correlation with Webb AI is 0.7021. Frey–Osborne remains an automation-risk comparator, not an AI architecture.

B Calendar, mapping, and sample flow

The original extract selected March ASEC (‘03s’) rather than March basic (‘03b’) for 2017–2021. ASEC records carry zero basic monthly final weights and cannot fill the intended cells. A revision extract restores the five basic samples. October 2025 remains absent because no CPS was collected during the federal government shutdown. Combined raw files contain 9,843,021 rows and 114 observed survey months; the corrected static model contains 113 after omitting December 2022.

TABLE 3. Unified sample flow

Universe	Occ.	Potential	Nonempty	Zero young	Zero older	Empty	Pre Y/O cov.	Post Y/O cov.
Characteristics	463	50,004	48,742	13,765	1,338	1,262	95.47/94.84	94.80/94.04
Beta + Webb	468	50,544	49,579	13,305	1,036	965	98.54/98.08	98.53/97.96
Beta + O*NET	465	50,220	48,909	13,698	1,385	1,311	95.39/95.39	94.02/94.52
Six + Webb	444	47,952	47,073	12,782	950	879	94.46/93.78	93.89/93.08

Notes: Frozen 108-month calendar. Coverage is percent of the 490-occupation candidate employment stock, separately for young (Y) and older (O) groups. Potential and nonempty columns are occupation-month cells. The four occupation counts reconcile earlier 463/468/465/444 universes.

Pre-2020 records use an official Census-2010-to-2018 route bridge. Conversion proportions are common to young and older observations within source occupation and month, mechanically preserving each source ratio across its components before aggregation. Fifty-six source occupations are one-to-many, maximum multiplicity is seven, and these sources represent 20.03 percent of early weighted employment.

The AIOE repair raises computer/mathematical employment coverage from 3.33 to 97.7 percent. A separate continuous decomposition is: original score/original support -0.01885 ;

TABLE 4. Calendar and taxonomy sensitivities

Specification	Months	Occ.	Coefficient	95% interval
Frozen calendar	108	468	-.1311	[-.2166, -.0455]
Restored March 2017–21	113	468	-.1346	[-.2230, -.0461]
Census-2018 period (2020+)	77	457	-.1211	[-.1986, -.0435]
Stable Census-2010 aggregate	113	408	-.1522	[-.2438, -.0606]
Respondent-equivalent cells	108	468	-.1308	[-.2149, -.0466]

Notes: Post-outcome exploratory. The post-2020 and stable-taxonomy rows change the temporal or occupational estimand. The respondent-equivalent row changes the cell outcome; it does not reweight weighted stocks.

repaired score/same 410 occupations -0.01920 ; repaired score/expanded 495 occupations -0.03156 ; expanded support excluding computer/math -0.02940 . Thus re-admission, not within-support score correction, supplies most of the difference.

C Residual-treatment and conditional-information support

Support depends on the statement being estimated. A continuous horse race consumes residual score variation after controls; a tail comparison consumes weighted group membership; a ranking statement consumes pairwise order. The diagnostics below are therefore not generic data-coverage statistics.

For the primary beta tail contrast, estimator information is concentrated more sharply than the nominal count suggests. The frozen information audit reports an effective occupation count of approximately 43.3 for the fitted Q5-versus-Q1 contrast, compared with 468 nominal occupations. The pre-outcome residual-treatment audit reports an effective count of 53.263. These objects have different weights and should not be equated.

The computerization comparison uses Webb’s software-exposure measure as the frozen primary comparison technology. Archived O*NET computer-use measures, routine-task intensity, and Frey–Osborne automation risk are sensitivities. Remotability is a separate proxy and not a measure of computerization. The comparison-technology results show that coefficient magnitude depends on the conditioning estimand; they do not select a uniquely correct exposure measure.

TABLE 5. Effective residual support across comparison technologies

Architecture	Webb	O*NET importance	O*NET level	RTI	Frey–Osborne
AIOE administrative	72.2	73.0	77.3	62.8	63.8
AIOE ability-direct	74.6	77.8	82.2	68.2	59.3
AIOE source-weighted	72.1	65.4	70.3	62.1	62.8
Alpha	17.4	31.1	31.7	11.9	26.3
Beta	53.3	63.2	59.5	51.7	62.9
Broad	84.5	44.4	72.2	77.5	74.0

Notes: Inverse-Herfindahl counts of employment-weighted residual-exposure variance after projecting each exposure on the named comparison technology. Frozen pre-outcome treatment-side analysis; no labor outcome enters.

Top-five residual-variance shares for the same columns are 18.4, 17.9, 16.6, 20.5, and 20.0 percent for administrative AIOE; 15.9, 16.5, 15.0, 17.8, and 20.6 for ability-direct AIOE; 18.1, 19.5, 18.3, 20.6, and 20.2 for source-weighted AIOE; 41.6, 32.8, 33.3, 46.6, and 36.0 for alpha; 22.2, 20.1, 20.6, 22.8, and 19.8 for beta; and 15.8, 25.2, 18.0, 17.3, and 17.1 for broad exposure.

D Estimator and inference audits

For two age groups, conditioning Poisson means (μ_y, μ_o) on their weighted total yields the grouped-binomial pseudo-log-likelihood $N_y \log p + N_o \log(1 - p)$ with $\text{logit}(p) = \log(\mu_y/\mu_o)$. Occupation-month effects cancel. CPS final weights build N_y and N_o and are not applied again to the objective. These continuous weighted totals define an estimating equation; they are not literal independent binomial trials. A multinomial age-profile extension requires its own derivation.

TABLE 6. Common-support sign inference and percent transformation

Measure	$\hat{\beta}$	One-sided p	Simult. upper	$100(e^\beta - 1)$	Upper, percent
AIOE administrative	-.0739	.045	.0186	-7.12	1.88
AIOE ability-direct	-.1029	.005	-.0167	-9.77	-1.66
AIOE source-weighted	-.1021	.006	-.0066	-9.71	-.66
Alpha	-.1013	.008	-.0070	-9.64	-.70
Beta	-.1290	.001	-.0269	-12.10	-2.65
Broad	-.1465	.001	-.0443	-13.63	-4.33

Notes: Identical 444-occupation support. The IUT p -value for all coefficients negative is the maximum marginal one-sided tail area, .045. The predeclared simultaneous critical value is 2.26036; the administrative upper bound is positive, so that separate criterion fails. Percent columns are monotone transformations of log relative-stock-ratio coefficients and limits.

The paired beta-minus-alpha contrast is -0.0324 (paired SE .0370, interval $[-.1023, .0376]$, $p = .403$). Common occupation multipliers preserve covariance. The interval permits only the design does not detect a difference, not economic equivalence.

Nominal primary clusters number 468. Webb-residualized beta has effective residual-variance count 53.26 and top-five share 22.15 percent; fitted conditional information has effective count 43.30 and top-five share 24.57 percent. Maximum fixed-treatment LOCO movement is .0205, with zero sign changes. These are distinct objects.

Two-way occupation/month inclusion–exclusion gives SE .0449 versus one-way .0444 and normal interval $[-.2191, -.0430]$. This is model-based, not a CPS survey-design correction. A literal 199-draw residual-wild full refit was attempted for the primary and no-alpha F/G models. Zero pseudo-outcomes were admissible because grouped-binomial successes left $[0, N]$; no clipping or alternative resampling was substituted.

E Employment-stock stress tests

All analyses in this section are post-outcome exploratory unless a row is explicitly called the frozen primary estimate. They retain the grouped employment-stock outcome and the primary age/timing/fixed-effect design as far as the stated contrast permits.

TABLE 7. Occupational-characteristic placebo benchmarks

Architecture	Top group	Coefficient	95% interval	AI minus benchmark
Eloundou beta	73	-.1142	[-.1981, -.0302]	—
Mean wage	50	-.0185	[-.0987, .0617]	-.0957
Education requirement	77	.0093	[-.0660, .0846]	-.1234
Cognitive content	56	.0270	[-.0416, .0955]	-.1411
Telework (zero/partial/full)	109	-.0240	[-.0837, .0357]	-.0901
STEM (binary)	29	-.0161	[-.1736, .1413]	-.0980

Notes: Identical 363-occupation support and common draws. AI-minus intervals exclude zero for education, cognition, and telework; intervals include zero for wage and STEM. Telework and STEM use natural groups because tie-preserving quintile cuts collapse.

Within-SOC2 permutation holds each broad family’s beta score multiset fixed and reassigns scores across detailed occupations. Among 999 successful fits, the coefficient mean is -0.1097 (SD .0268); the observed -0.1311 has lower-tail probability .219 and centered two-sided probability .434. Exchangeability is not established, so this is an assumption-dependent diagnostic.

TABLE 8. Reference-tail contrasts

Contrast	Coefficient	SE	95% interval
Q5–Q1	-.1311	.0442	[-.2179, -.0443]
Q5–Q2	-.0456	.0397	[-.1240, .0328]
Q5–Q3	-.0833	.0387	[-.1582, -.0083]
Q5–Q4	-.0340	.0393	[-.1141, .0460]
Q4–Q2	-.0116	.0310	[-.0725, .0494]

Notes: Re-referencing uses the fitted primary model and common draws; it does not change the sample or treatment. The Q3-relative coefficients are Q1 .0478, Q2 $-.0377$, Q3 0, Q4 $-.0492$, and Q5 $-.0833$.

In a separate six-bin multinomial age profile relative to 51–65, coefficients are $-.0744$ (18–21), $-.0996$ (22–25), .0150 (26–30), .0645 (31–40), and .0344 (41–50). Only the 31–40

TABLE 9. Age-comparison estimates

Comparison	Coefficient	Paired SE	95% interval
22–25 vs 26–65	-.1311	.0443	[-.2190,-.0431]
22–25 vs 26–35	-.1307	.0334	[-.1945,-.0669]
22–25 vs 26–45	-.1475	.0389	[-.2240,-.0709]
22–25 vs 36–55	-.1464	.0499	[-.2438,-.0489]
22–25 vs 51–65	-.0978	.0527	[-.2000,.0045]
18–21 vs 51–65	-.1050	.0618	[-.2251,.0151]

Notes: Relative to 22–25 versus 26–65, paired differences are .0003, $-.0164$, $-.0153$, and .0333 for the next four 22–25 comparisons. Only the final interval excludes zero.

interval excludes zero. This profile is not an additive decomposition of the binary pooled comparison.

TABLE 10. Exclusions, cell-size, and remote-work sensitivities

Specification	Occ.	Coefficient	95% interval
Exclude food-preparation/serving SOC35	458	-.1237	[-.2039,-.0434]
Exclude computer/math SOC15	453	-.1158	[-.2065,-.0251]
Exclude office/admin SOC43	421	-.1163	[-.1951,-.0375]
Exclude approximate IND1990 leisure/hospitality, fixed membership	467	-.1325	[-.2191,-.0460]
Same, recomputed quintiles	467	-.1388	[-.2286,-.0491]
Minimum respondent-equivalent 100	463	-.1311	[-.2198,-.0423]
Minimum respondent-equivalent 250	461	-.1311	[-.2180,-.0441]
Minimum respondent-equivalent 500	438	-.1326	[-.2181,-.0470]
Beta + remote, AI coefficient	408	-.1077	[-.1934,-.0220]
Beta + remote, remote coefficient	408	.0065	[-.0214,.0343]
Alpha + remote, AI coefficient	408	-.0808	[-.1614,-.0003]
Alpha + remote, remote coefficient	408	-.0132	[-.0371,.0108]

Notes: The IND1990 approximation includes code 800, which bundles theaters and motion pictures. Remote coefficients are per weighted SD of Dingel–Neiman feasibility; quintiles are recomputed on joint support. These models do not identify an AI-versus-remote causal decomposition.

External architectures estimate -0.0649 for Webb AI (448 occupations, interval $[-.1439, .0141]$) and -0.0110 for the reversed OECD capability gap (448, interval $[-.0864, .0645]$). On common support, their differences from beta are $.0646$ (interval $[-.0411, .1704]$) and $.1115$ (interval $[-.0015, .2244]$). The six selected-family sign result is therefore architecture-set dependent.

F Dynamics and time heterogeneity

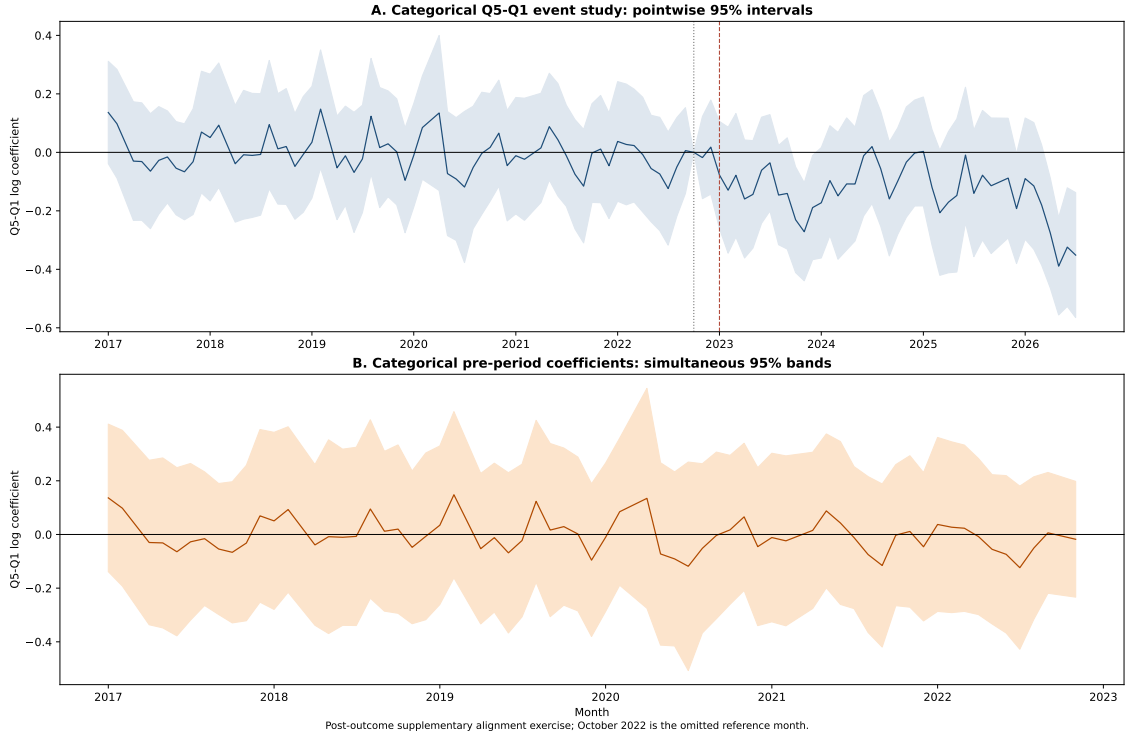


FIGURE 2. Categorical beta Q5–Q1 event-time profile. The plotted object is the fitted tail contrast, not mapping coverage or continuous residual information. December 2022 is the omitted transition month; October 2025 is absent because the survey was not collected.

The preperiod max-statistic is 1.502 with common-multiplier $p = .929$, so the implemented event study does not detect a differential pretrend. This does not establish parallel trends or attribute a later break to AI. Forty of 42 observed post coefficients are negative, but the path is neither immediate nor monotone.

The post-2025 extension is different from 2024 but not from 2023. Interest rates, technology-sector adjustment, return-to-office changes, post-pandemic normalization, and CPS composition remain concurrent explanations. The failed 5,148-parameter occupation-age seasonality model is recorded rather than replaced.

TABLE 11. Joint post-period coefficients

Period or contrast	Coefficient	Paired SE	95% interval
2023	-.1356	.0452	[-.2245,-.0467]
2024	-.0783	.0548	[-.1821,.0254]
2025–26	-.1629	.0507	[-.2623,-.0635]
2024 minus 2023	.0573	.0447	[-.0302,.1447]
2025–26 minus 2023	-.0273	.0440	[-.1154,.0608]
2025–26 minus 2024	-.0846	.0374	[-.1577,-.0115]

G Longitudinal flow estimands and results

All flow models use valid exact adjacent-month CPSIDV links, origin age, official origin LNKFW1MWT, January 2017–November 2022 as pre, January 2023 onward as post, and December 2022 as transition. Links across the MISH 4-to-5 rotation gap are excluded. December 2019 origins are excluded from occupation-dependent transitions.

For exit, the risk set is employed origins and the event is nonemployment at $t + 1$. For occupational outflow, the risk set is employed origins linked to an employed destination and the event is a different harmonized occupation. For entry destination, the risk set is nonemployed origins linked to employment; the outcome is allocation toward destination Q5 rather than Q1. Consequently, the three coefficients are not additive decompositions of the stock coefficient.

TABLE 12. Longitudinal flow estimates

Margin	Coefficient	Analytic SE	Wild CI	p
Employment exit	0.1195	0.0944	$[-0.0668, 0.3059]$	.229
Occupational outflow	0.0107	0.0631	$[-0.1063, 0.1277]$	.857
Persistent outflow	−0.0246	0.1195	$[-0.2580, 0.2088]$	.850
Entry destination	−0.0888	0.0976	$[-0.2787, 0.1011]$	.383

Notes: Post-outcome exploratory conditional-Poisson rate or allocation models using the frozen beta Q5-versus-Q1 contrast and official longitudinal weights. Inference uses 999 occupation-cluster wild-score draws. Each interval includes zero. Entry destination is an allocation estimand, not an employment-finding probability.

Dependent interviewing does not eliminate occupation coding error. The analysis reports a 9.865 percent immediate-reversal rate and repeats the switch decomposition among legitimate $A \rightarrow B \rightarrow B$ observations. That sensitivity conditions on observing a third adjacent interview and is not a general correction for coding error.

H Mobility thresholds and rematching

Zero-threshold conflict is 35.02 percent for the five-measure no-alpha set and 11.35 percent within AIOE; movement-mass-weighted shares are 15.35 and 2.10 percent. At .25 SD, substantial opposition is 12.77 percent without alpha and .33 percent within AIOE. These comparisons change the number of potential pairs and are not marginal effects of deleting a measure.

TABLE 13. Six-way mobility conflict by standardized threshold

Threshold	Eligible weight	Any conflict eligible	Opposition/all	Opposition eligible
0	100.00	53.28	53.28	53.28
.10	99.78	53.18	43.35	43.45
.25	97.42	52.78	29.79	30.58
.50	85.29	50.99	14.00	16.41

Notes: Percent, official longitudinal weights, 108,500 switches. “Any conflict” requires one positive and one negative sign after restricting to switches where at least one architecture exceeds the threshold. “Opposition” requires both directions to exceed it. Movement-mass-weighted conflict is 40.10 percent at all thresholds because its weights, rather than its eligible set, use total absolute movement.

Immediate reversals (6,187 switches) and persistent A–B–B changes (39,893) have zero-threshold conflicts of 55.15 and 54.46 percent. At .5 SD, substantial-opposition shares are 13.57 and 14.39 percent. Transition-boundary exclusions are applied symmetrically; no September-to-November 2025 false adjacent links and no MISH4-to-5 long-gap pairs remain.

Across architectures, weighted shares moving higher range from 49.21 to 50.49 percent; destination top-rank-quintile shares range from 19.96 to 23.39 percent. The switch-only frame cannot classify entrants from nonemployment.

TABLE 14. Hard-rematching benchmark audit

Object	50,000 pseudo	100,000 pseudo	200,000 pseudo
Draws	199	199	999
Represented weight	73.86%	95.30%	98.31%
Benchmark conflict	52.3588%	52.3718%	52.3227%
Realized minus benchmark	0.9240 pp	0.9110 pp	0.9601 pp
Monte Carlo SE of mean	0.0078 pp	0.0054 pp	0.0017 pp

The main benchmark uses 30,170 strata and preserves detailed origin/destination margins within age-by-month and broad origin/destination cells. A no-self repair eliminates all generated identical detailed origins and destinations while preserving margins. The exact unconstrained product-marginal conflict is 50.2893 percent across 32,075 strata but permits self-transitions with probability 3.6591 percent; it is not the exact expectation for the constrained null. The 0.96-point excess is a sample finding close to the predeclared one-point threshold, not an equivalence conclusion.

Named threshold-surviving examples include accountants to bookkeeping clerks; physician assistants to diagnostic technologists; registered nurses to medical and health-services managers; and retail salespersons to cashiers. Complete score movements and weights are stored in the machine-readable results.

I Continuous family models and exact reparameterization

Let A and E be equal-weight centroids of the three standardized ability- and task-family implementations. Raw $F = (A + E)/2$ and $G = (A - E)/2$ are separately standardized in the fitted model with Webb. The original construction has weighted SDs .8777 and .3221 and weighted correlation .2177; neither raw construction nor standardization makes the fitted coordinates orthogonal.

TABLE 15. Leave-one-measure F/G outcome refits

Omitted measure	b_F	95% interval	b_G	95% interval
None (original)	-.0404	[-.0674,-.0133]	.0309	[.0029,.0589]
Alpha	-.0330	[-.0571,-.0088]	.0342	[.0089,.0596]
AIOE administrative	-.0404	[-.0676,-.0133]	.0294	[.0009,.0580]
AIOE ability-direct	-.0401	[-.0672,-.0129]	.0338	[.0059,.0618]
AIOE source-weighted	-.0404	[-.0676,-.0133]	.0293	[.0015,.0572]
Beta	-.0421	[-.0702,-.0141]	.0287	[-.0009,.0584]
Broad	-.0383	[-.0652,-.0114]	.0215	[-.0080,.0511]

Notes: Identical 444-occupation support and family rebalancing after each omission. Each row refits the labor outcome with 999 common occupation multipliers. Results are post-outcome exploratory.

A representative two-score model estimates administrative AIOE .0449 (interval [.0012,.0885]) and beta $-.0788$ (interval $[-.1292, -.0284]$). The direct primitive model estimates $D = -.0323$ (interval $[-.0617, -.0030]$) and $S = \gamma - \alpha = -.0303$ (interval $[-.0562, -.0044]$), conditional on Webb.

For separately scaled regressors, the exact basis transformation is

$$\gamma_A = \frac{1}{2}(b_F/s_F + b_G/s_G), \quad \gamma_E = \frac{1}{2}(b_F/s_F - b_G/s_G).$$

Consistently transforming every draw gives $\gamma_A = .0248$ (paired SE .0199) and $\gamma_E = -.0707$ (paired SE .0272). This is the same fitted predictor, not corroboration. A smaller variance alone does not mechanically create a larger standardized coefficient; covariance with the residualized outcome and with other regressors also matters.

For $D + \lambda S$, the categorical Q5–Q1 estimate is $-.1013$ at $\lambda = 0$, $-.1256$ at .25, $-.1297$ at .5, $-.1432$ at .75, and $-.1482$ at 1. Continuous slopes differ under fixed beta scaling and restandardization, so both conventions are stored. The combined evidence supports construction-sensitive asymmetry, not a universal family ordering.

J Fixed-treatment occupation influence

The leave-one-occupation-out audit deletes each occupation while retaining frozen treatment assignments and refits the same model. It is an influence diagnostic, not a redefinition of treatment. The primary full-sample estimate is -0.131074 ; deletion estimates range from -0.14238 to -0.11055 , with no sign reversal. The largest absolute movement is 0.02052 .

Fast food and counter workers are in frozen Q1; deleting them moves the estimate to -0.110553 (movement $+0.020521$). Customer service representatives are in Q5; deleting them moves the estimate to -0.113112 (movement $+0.017962$). Software developers are in Q5 and rank tenth by absolute influence; deleting them moves the estimate to -0.138787 (movement -0.007713). Thus, the result is not a software-developer artifact, while influential contributions arise at both exposure tails.

TABLE 16. Ten largest absolute occupation-deletion movements

Rank	Deleted occupation	Delete estimate	Movement
1	Fast food and counter workers	-0.11055	$+0.02052$
2	Customer service representatives	-0.11311	$+0.01796$
3	Home health aides	-0.14238	-0.01131
4	Computer occupations, all other	-0.12080	$+0.01028$
5	Office clerks, general	-0.14109	-0.01002
6	Secretaries and administrative assistants ^a	-0.14089	-0.00982
7	Food preparation workers	-0.13981	-0.00874
8	Laborers and freight/material movers, hand	-0.13946	-0.00839
9	Wholesale and manufacturing sales representatives	-0.13924	-0.00817
10	Software developers	-0.13879	-0.00771

Notes: Post-outcome fixed-treatment influence audit. The complete machine-readable table contains all 468 deletion estimates. Employment-stock-weighted signed-movement percentiles (1st, 5th, median, 95th, 99th) are -0.01002 , -0.00817 , -0.00001 , 0.00377 , and 0.01796 .

a Except legal, medical, and executive.

For the family-disagreement coefficient, all 444 deletion estimates remain positive between 0.025128 and 0.035993 . The ten largest absolute movements follow deletion of driver/sales workers and truck drivers (-0.005766), bookkeeping/accounting/auditing clerks ($+0.005100$), computer programmers ($+0.004516$), couriers and messengers (-0.004237), fast food and counter workers (-0.003463), insurance claims and policy-processing clerks (-0.003221), payroll and timekeeping clerks ($+0.002869$), home health aides ($+0.002618$), sales managers ($+0.002605$), and other computer occupations (-0.002509). Employment-stock-weighted signed-movement percentiles (1st, 5th, median, 95th, and 99th) are -0.005766 , -0.001693 , -0.000024 , 0.002155 , and 0.004516 .

K Prospective precision and failed diagnostics

The prospective primary SE was .01217 and realized SE .04441, a ratio of 3.649. For beta minus alpha, .01167 versus .03697 gives 3.167. Estimands, support, units, and inference targets match. The ex-ante paired design had 80-percent power to detect differences of approximately 3.27 percentage points, but no verified literature-comparable smallest effect of interest existed; binding equivalence inference was retired rather than assigned an arbitrary threshold.

The historical simulation preserved observed exposure, weights, design matrices, and ordered residual paths. It did not preserve contemporaneous cross-occupation covariance and did not include survey noise, new post shocks, structural composition change, exposure measurement error, or misspecification. Available variants do not identify their separate contributions. Comparing $\sqrt{468/43.3}$ to the precision ratio is only a scale heuristic because 43.3 is fitted information, not independent clusters or the simulated information target.

Two failed diagnostics remain part of the record. The occupation-specific age-season model did not converge with 5,148 additional nuisance parameters. The requested literal residual-wild full refit yielded zero admissible pseudo-outcomes in 199 attempts for both the primary and no-alpha F/G models because successes left grouped-binomial bounds. No clipping, alternative bootstrap, or favorable substitute was introduced.

L Reproducibility and design chronology

The design-freeze tag resolves to commit 22fbf7924809b7a535e31ae0ab68f5b113ce8078; the confirmatory-results tag resolves to b16109482c3bf5ca176f6f08976e120b04769945. Frozen artifacts are unchanged. This referee revision lives on a separate branch; calendar repair, placebos, age and era comparisons, external architectures, leave-one-measure family fits, threshold mobility, inference audits, and manuscript changes are post-outcome exploratory.

Public-use CPS microdata are obtained from IPUMS and are not redistributed. The four principal input hashes are listed below; external public inputs and every result file are separately hashed in receipts.

Original CPS file (SHA-256):

3fe42477e6f2ce401e85123f0e278e758595c1c4071a8743f243a92752db38c9

March-basic repair:

a4ae2ef06d66a0d47359ccceffff9a15843ac501a9f25f9a05aa8fdca9c4f911

Occupation bridge:

0bd2f63c72e24bed2cc1cb414395c3cbddf7c00011e47ec1c1de6ae534fd1dcc

Exposure lookup:

`c6eb70623ea598bfc41f2352391add7a342a8809a4e57b02f2af0e75dd0223f8`

The documented execution order is: core stress tests; rebuilt-cell and sample-flow audits; mobility thresholds and hard benchmark; external architectures; inference audits; figures; LaTeX production. Scripts, seeds, draw counts, support hashes, software paths, and output provenance accompany the machine-readable results. Private credentials, raw restricted paths, and authentication files are excluded from the package.

Author affiliation, email, acknowledgments, funding, and disclosure text remain marked for author completion rather than invented.